

Chapter 5: Time and Causality (5 hrs)

1. Times Series Analysis

- a) Autocorrelation and stationary
- b) ARMA and ARIMA models
- c) Selecting optimal lag length: Akaike and Bayesian Information criteria

2. Causal Analysis

- a) Causation vs correlation
- b) Granger causality: overview only
- c) Causal Directed Acyclic Graph: overview only

5.1 Times Series Analysis

What is a Time Series?

A **time series** is a set of observations of a variable **collected over time** in a **specific order** (usually at equal time intervals).

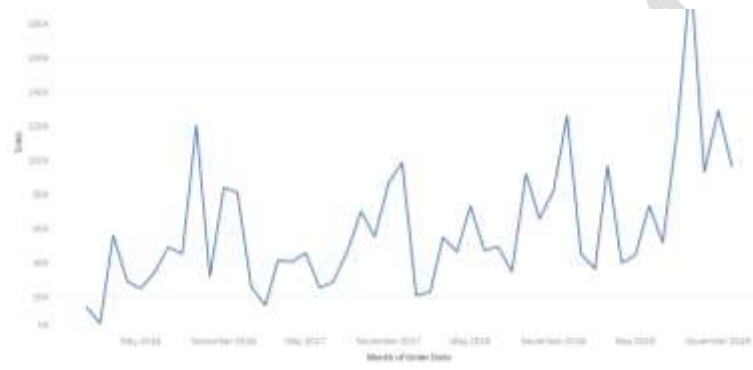
A time series is a sequence of data points indexed in time order.

Mathematically,

$$\{X_t\} = X_1, X_2, X_3, \dots, X_t$$

Where:

- t = time (day, month, year, etc.)
- X_t = value at time t



Examples

Domain	Time Series Example
Economics	Annual GDP, inflation rate
Finance	Stock prices, exchange rates
Weather	Daily temperature
Healthcare	Heartbeat signal
Engineering	Sensor readings

Characteristics of Time Series Data

Time series data has **time dependency**, meaning:

- Order of data matters
- Observations are often correlated with past values

Unlike cross-sectional data, time series data **cannot be randomly shuffled**.

5. Time and Causality

Time Series Analysis:

- Time series analysis is a specific way of analyzing a sequence of data points collected over an interval of time.
- Time series analysis involves **studying past data points** to understand patterns and **forecast future values**.
- In time series analysis, analysts record data points at consistent intervals over a set period of time rather than just recording the data points intermittently or randomly
- In general, there are two goals in time series analysis.
 - *Modeling Time Series*: Generating the time series with underlying mechanism.
 - *Forecasting Time Series*: Predict the future values of the time series variables.
- Time series data mining is the process of extracting patterns, trends, and relationships from time series data.
- It involves visualizing the data, decomposing the different components, and using statistical and machine learning techniques to analyze the data and make predictions about future trends.
- An example of time series data mining could be analyzing hourly electricity consumption data to identify daily and seasonal patterns, decomposing the data to separate out the trend and seasonality, and using statistical or machine learning techniques to make predictions about future consumption values.

Application of Time Series Analysis:

1.	Financial:	• Used for stock price evaluation • For the measurement of Inflation
2.	Industry:	• Determine the power consumption
3.	Scientific:	• Used for experiment results
4.	Meteorological:	• Concerned with the processes and phenomena of the atmosphere, basically for forecasting weather

Objectives of Time Series Analysis

1. Identify patterns in data
2. Understand data behavior
3. Forecast future values
4. Support decision making
5. Detect anomalies

Steps in Time Series Analysis

Step 1: Data Collection

- Data must be equally spaced
- Missing values should be handled

Step 2: Visualization

- Plot time vs value
- Helps identify trend and seasonality

Step 3: Stationarity Check

- Check if mean and variance are constant
- Use plots or statistical tests

Step 4: Model Selection

- AR, MA, ARMA, or ARIMA models
- Choose lag length

Step 5: Parameter Estimation

- Estimate coefficients using data

Step 6: Model Evaluation

- Check residuals
- Use AIC/BIC

Step 7: Forecasting

- Predict future values

Types of Time Series

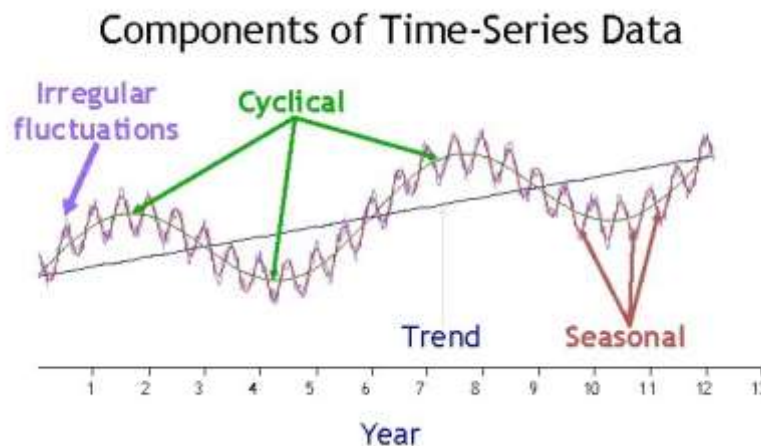
(a) Univariate Time Series

- One variable over time
- **Example:** Daily rainfall

(b) Multivariate Time Series

- Multiple related variables over time
- **Example:** Temperature, humidity, pressure

Components of time-series movements (Trend analysis)



There are three main categories of time series movement:

1. **Trend:** The trend component represents the long-term movement of the time series data in a particular direction. Trends can be upward (increasing over time), downward (decreasing over time), or stationary (remaining constant over time).
2. **Seasonality:** The seasonality component represents the pattern of the time series that repeats itself at regular intervals, such as daily, weekly, or monthly. Seasonality is typically associated with calendar events, holidays, or weather patterns.
3. **Cyclical:** The cyclical component represents the fluctuations in the time series that do not have a fixed periodicity. These fluctuations are often influenced by economic, political, or other external factors that affect the overall movement of the time series. For example, business cycles.
4. **Irregular or random movements:**

These fluctuations are unforeseen, uncontrollable and unpredictable. They are not regular variations and are purely random or irregular.

a) Autocorrelation and stationary

Autocorrelation:

Autocorrelation measures the **relationship between a variable and its past values**.

- It shows how current value depends on previous values
- Used to detect patterns in time series
- Autocorrelation value lies between **-1 and +1**
- Positive → values move in same direction
- Negative → values alternate

Example

If today's temperature is high, yesterday's temperature was probably high too
→ This shows **positive autocorrelation**.

Autocorrelation Function (ACF)

Correlation between X_t and X_{t-k} , where k is the lag.

$$\rho_k = \frac{\text{Cov}(X_t, X_{t-k})}{\text{Var}(X_t)}$$

Interpretation:

- High autocorrelation → strong dependence on past values
- Near zero → weak or no dependence

Types of Autocorrelation

Type	Description
Positive	High values followed by high values
Negative	High values followed by low values
Zero	No relationship

Problem with Autocorrelation

- Violates independence assumption
- Can lead to misleading regression results

5. Time and Causality

Numerical:

Consider a simple time series with 5 observations as below. Compute autocorrelation at lag-1.

Time (t)	Value X_t
1	2
2	4
3	6
4	8
5	10

Solution:

Step 1: Compute the Mean

$$\bar{X} = \frac{2 + 4 + 6 + 8 + 10}{5} = \frac{30}{5} = 6$$

Step 2: Create Lag-1 Series

X_t	X_{t-1}
4	2
6	4
8	6
10	8

Step 3: Compute Deviations from Mean and their product.

X_t	$X_t - \bar{X}$	$X_{t-1} - \bar{X}$	$(X_t - \bar{X})(X_{t-1} - \bar{X})$
4	-2	-4	$(-2)(-4) = 8$
6	0	-2	$(0)(-2) = 0$
8	2	0	$(2)(0) = 0$
10	4	2	$(4)(2) = 8$

$$\Sigma = 16$$

Step 4: Compute Variance Term

$$\sum (X_t - \bar{X})^2$$

X_t	$X_t - \bar{X}$	$(X_t - \bar{X})^2$
2	-4	16
4	-2	4
6	0	0
8	2	4
10	4	16

$$\Sigma=40$$

Step 5: Autocorrelation Formula

$$\rho_1 = \frac{\sum (X_t - \bar{X})(X_{t-1} - \bar{X})}{\sum (X_t - \bar{X})^2}$$

$$\rho_1 = \frac{16}{40} = 0.4$$

Final Answer $\rho_1 = 0.4$

Interpretation

- The autocorrelation at **lag-1 is positive**
- Indicates **moderate dependence** on previous value
- Suitable for AR-type models.

Assignment:

Consider a simple time series with 5 observations as in above example. Compute autocorrelation at lag-2.

Stationarity:

A time series is **stationary** if its statistical properties **do not change over time**.

Example:

- Stationary: daily fluctuations around a fixed mean
- Non-stationary: steadily increasing population

Why Stationarity is Important?

- Most time series models (ARMA, ARIMA) assume stationarity
- Non-stationary data can give misleading results

Types of Stationarity:

(a) Strict Stationarity

- Entire distribution does not change over time
- Very strong condition
- Rarely used in practice

(b) Weak (Second-Order) Stationarity

- Mean, variance, covariance are constant
- Most commonly used

Conditions for Stationarity:

A time series is stationary if:

1. Mean is constant

$$E(X_t) = \mu$$

2. Variance is constant

$$\text{Var}(X_t) = \sigma^2$$

3. Autocovariance depends only on lag, not time

$$\text{Cov}(X_t, X_{t-k}) = \gamma_k$$

Numerical:

Consider the following time series data:

Time (t)	X _t
1	8
2	10
3	9
4	11
5	10
6	9

- (a) Calculate the mean of the given time series.
- (b) Calculate the variance of the series.
- (c) By dividing the data into two equal parts, comment on whether the mean remains constant over time.
- (d) Based on your results, determine whether the time series is stationary or non-stationary. Give reasons.

Solution:

Step 1: Check Mean

$$\bar{X} = \frac{8 + 10 + 9 + 11 + 10 + 9}{6} = \frac{57}{6} = 9.5$$

Split data into two parts to see if mean changes.

- First half: 8,10,9
Mean = 9
- Second half: 11,10,9
Mean = 10

Mean is approximately constant.

Step 2: Check Variance

Compute deviations from mean:

X_t	$X_t - 9.5$	$(X_t - 9.5)^2$
8	-1.5	2.25
10	0.5	0.25
9	-0.5	0.25
11	1.5	2.25
10	0.5	0.25
9	-0.5	0.25

$$\text{Variance} = \frac{5.5}{6} \approx 0.92$$

Variance remains stable across time.

Step 3: Autocovariance Depends on Lag

- Relationship between X_t and X_{t-1} is similar throughout
- No visible trend or seasonality

Conclusion

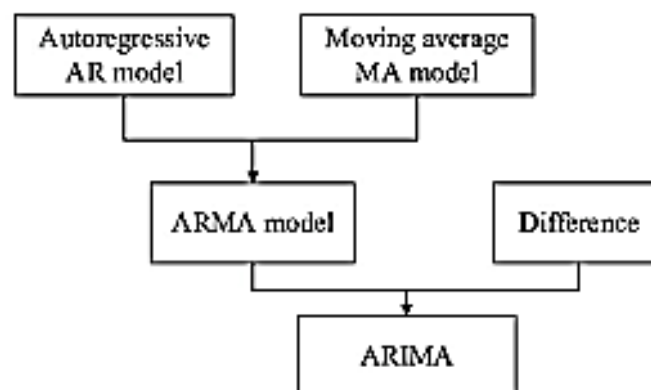
Mean $\approx$ constant

Variance $\approx$ constant

Autocovariance depends only on lag

This time series is stationary

b) ARMA and ARIMA models



ARMA (Autoregressive Moving Average) model

Time series analysis is a crucial aspect of data science, particularly when dealing with data that is collected over time. One of the fundamental models used in time series analysis is the ARMA (Autoregressive Moving Average) model

The ARMA model is a combination of two simpler models: the Autoregressive (AR) model and the Moving Average (MA) model. The ARMA model is used to describe time series data that is stationary, meaning its statistical properties do not change over time.

The **ARMA model** combines two parts:

a) AR (Autoregressive) part

- The current value depends on its **own past values**.
- **Example:** Today's temperature depends on the temperature of the last 2 days.
- AR model of order p (AR(p)):

$$X_t = \phi_1 X_{t-1} + \phi_2 X_{t-2} + \dots + \phi_p X_{t-p} + \epsilon_t$$

Where:

- X_t = current value
 - $\phi_1, \phi_2, \dots, \phi_p$ = coefficients
 - ϵ_t = white noise/error term
 - p = number of lag observations used
- **Example:** Suppose we predict stock price:

$$X_t = 0.7X_{t-1} + 0.2X_{t-2} + \epsilon_t$$

Yesterday's price counts more (0.7), day before counts less (0.2).

b) MA (Moving Average) part

- The current value depends on **past errors/noise**.
- **Example:** Today's stock price is influenced by yesterday's forecast error.
- MA model of order q (MA(q)):

$$X_t = \epsilon_t + \theta_1 \epsilon_{t-1} + \theta_2 \epsilon_{t-2} + \dots + \theta_q \epsilon_{t-q}$$

5. Time and Causality

Where:

- ϵ_t = current error term
- $\theta_1, \theta_2, \dots, \theta_q$ = coefficients
- q = number of lagged forecast errors

ARMA (p,q) model

Combines AR(p) and MA(q):

$$X_t = \phi_1 X_{t-1} + \dots + \phi_p X_{t-p} + \epsilon_t + \theta_1 \epsilon_{t-1} + \dots + \theta_q \epsilon_{t-q}$$

- p = number of AR terms (past values)
- q = number of MA terms (past errors)

Analogy:

- Predicting your mood today:
 - AR: influenced by how you felt yesterday and day before.
 - MA: influenced by how wrong yesterday's mood prediction was.

Assumption: Time series is **stationary** (its mean and variance don't change over time).

Simple Example

Suppose daily sales of a shop are roughly stationary (around 100 units/day). We notice:

- Yesterday's sales influence today's $\rightarrow$ AR(1)
- Yesterday's prediction error influences today $\rightarrow$ MA(1)

ARMA(1,1) formula:

$$X_t = 0.6X_{t-1} + \epsilon_t + 0.3\epsilon_{t-1}$$

- Yesterday we sold 120 units
- Yesterday's error was +10 (we predicted 110, sold 120)
- Today's prediction:

$$X_t \approx 0.6 \times 120 + 10 + 0.3 \times 10 = 72 + 10 + 3 = 85$$

So we predict around 85 units for today.

ARIMA (Autoregressive Integrated Moving Average) model

ARIMA is a **model used to predict future values** in a series of data points over time (time series).

It is useful when the data has **trends** or changes over time.

ARIMA stands for:

- **AR** → **AutoRegressive**: Past values influence the current value.
- **I** → **Integrated**: Differencing of data to make it stationary.
- **MA** → **Moving Average**: Past forecast errors influence the current value.

So, ARIMA is a **time series forecasting model** that works well when your data shows trends (non-stationary).

Why “Integrated”?

- Many real-world time series (like stock prices, temperature, or sales) are **non-stationary**, meaning their mean or variance changes over time.
- **Differencing** is used to make it stationary:

$$Y'_t = Y_t - Y_{t-1} \quad (1\text{st difference})$$

- You can difference more than once if needed (2nd difference, etc.).
- The number of differences used is the “**I**” in ARIMA, usually denoted as **d**.

Example in words:

If your monthly sales are increasing over time:

- Subtract last month's sales from this month's (d=1).
- Use the previous month's sales (p=1) and last month's error (q=1).
- This gives ARIMA(1,1,1), which can forecast next month's sales.

ARIMA(p, d, q)

- p = order of AR part
- d = number of differences needed to make the series stationary
- q = order of MA part (how many past forecast errors we use).

Mathematically:

$$(1 - \phi_1 L - \phi_2 L^2 - \dots - \phi_p L^p)(1 - L)^d Y_t = (1 + \theta_1 L + \theta_2 L^2 + \dots + \theta_q L^q) \epsilon_t$$

Where:

- L = lag operator ($LY_t = Y_{t-1}$)
- ϕ_i = AR coefficients
- θ_i = MA coefficients
- ϵ_t = white noise

Steps to use ARIMA

1. **Check the data** → See if it is stationary (stable).
2. **Make it stationary** → If not, subtract previous values (differencing).
3. **Select AR and MA values** → Use past data patterns to choose how many past values (AR) and past errors (MA) to include.
4. **Build the model** → Fit the ARIMA model with the selected values.
5. **Forecast** → Predict future values using the model.

c) Selecting optimal lag length: Akaike and Bayesian Information criteria

- In **time series models** (like AR, MA, ARMA, ARIMA), the **lag length** tells us how many **past values** (or past errors) to include in the model.
- Choosing the **right number of lags** is important:
 - Too few → Model misses patterns → Poor forecast.
 - Too many → Model too complex → Overfitting.

Steps to select optimal lag:

1. Fit models with **different lag lengths** (p, q).
2. Calculate **AIC and BIC** for each model.
3. Choose the lag where **AIC or BIC is minimum**.

Usually, BIC gives a **simpler model**, while AIC may include slightly more lags.

Information criteria for selecting lags:

i) Akaike Information Criterion (AIC)

Formula:

$$AIC = -2 \cdot \ln(L) + 2k$$

Where:

5. Time and Causality

- L = likelihood of the model (how well it fits the data)
- k = number of parameters (lags)
- **Goal:** Choose the model with **lowest AIC**.
- Balances **good fit** and **simplicity**.

ii) Bayesian Information Criterion (BIC)

Formula:

$$BIC = -2 \cdot \ln(L) + k \cdot \ln(n)$$

Where:

- n = number of observations
- k = number of parameters (lags)
- **Goal:** Choose the model with **lowest BIC**.
- Penalizes complexity **more heavily than AIC**

5.2 Causal Analysis

Causal analysis is the process of **studying and identifying cause-and-effect relationships** between variables. It goes beyond just observing patterns or correlations—it tries to answer:

“Does X actually cause Y ?”

Simple Example:

- Suppose you notice that students who study more hours tend to get higher exam scores.
- **Causal analysis** would try to determine:
 - Does studying more **actually cause** higher scores?
 - Or is there something else (like prior knowledge or tutoring) influencing the scores?

a) Causation vs correlation

Correlation

- **Definition:** A statistical measure that shows **how two variables move together**.
- **Key Point:** Correlation **does not mean one causes the other**.
- **Values:**
 - Positive correlation → Both variables increase or decrease together.
 - Negative correlation → One increases while the other decreases.
- **Example:** Ice cream sales increases and drowning deaths increases in summer.
 - They are correlated, but eating ice cream does **not cause** drowning.
 - The hidden factor is **hot weather**.

Causation

- **Definition:** When a change in one variable **directly causes** a change in another variable.
- **Key Point:** Causation implies a **cause-effect relationship**.
- **Example:** Smoking → Lung cancer.
 - Smoking directly increases the risk of lung cancer.

5. Time and Causality

Differences:

Aspect	Correlation	Causation
Definition	A statistical relationship between two variables	One variable directly affects or changes another
Direction	Can be positive or negative	Always directional (cause → effect)
Implication	Does not imply one causes the other	Implies that changing the cause changes the effect
Example	Ice cream sales and drowning deaths are correlated (both increase in summer)	Smoking causes an increase in lung cancer risk

- **Correlation is easier to detect** using statistical data.
- **Causation requires deeper analysis** (experiments, interventions, or causal models).

b) Granger causality: overview only

Granger causality is a statistical concept used in **time series analysis** to test whether **one variable can help predict another variable**.

- Granger causality **does not prove true causation**. It only shows predictive causality.
- **Clive Granger** introduced this method in **1969–1980** for analyzing time series data.
- He developed a statistical test to check whether one time series can **predict another**.
- For this work, he shared the **2003 Nobel Prize in Economics** for contributions to analyzing **economic time series**.

So, it's called **Granger causality** because Granger invented the idea of testing “predictive causality” in time series.

Key Idea: If knowing the **past values of X** helps predict the **future values of Y** better than only using past values of Y, then **X Granger-causes Y**.

Simple Example

Scenario:

- X = Monthly advertising spending

5. Time and Causality

- Y = Monthly sales

Observation:

- We want to see if advertising spending helps predict sales.

Analysis:

1. Predict sales (Y) using only past sales $\rightarrow$ Model A.
2. Predict sales using past sales **and** past advertising spending $\rightarrow$ Model B.
3. Compare the models:
 - If Model B predicts sales better $\rightarrow$ Advertising **Granger-causes** sales.

Interpretation:

- This means advertising spending is **useful for predicting future sales**, even though it does not prove that spending directly causes sales to rise.

c) Causal Directed Acyclic Graph (DAG): overview only

A Causal DAG is a **visual representation of cause-and-effect relationships** using:

- **Nodes:** variables (things we measure)
- **Arrows:** direction of causal effect (cause $\rightarrow$ effect)

Acyclic:

- “Acyclic” means there are **no loops**.
- A variable cannot indirectly cause itself.

Purpose:

- Helps identify **causal pathways**.
- Helps spot **confounders** (variables affecting both cause and effect).
- Useful in experiments or observational studies.

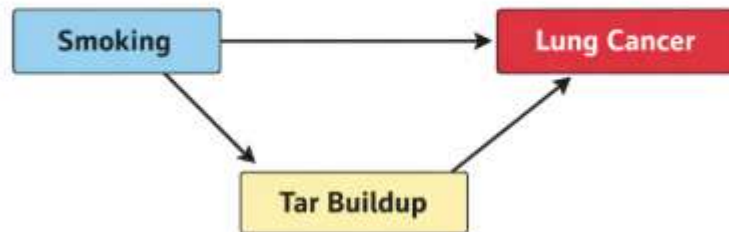
Simple Example:

Scenario:

- **Smoking** can cause **Tar buildup** in lungs, which in turn can cause **Lung Cancer**.
- **Smoking** also directly affects **Lung Cancer**.

DAG Representation:

Causal Directed Acyclic Graph (DAG)



Explanation:

- Smoking → Lung Cancer (direct effect)
- Smoking → Tar Buildup → Lung Cancer (indirect effect)
- The arrows show **causal direction**.
- No cycles: nothing loops back to Smoking.
